

Jiayi Li

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PROFILE	I study what can be known about a trained neural network. Using algebraic geometry, singular learning theory, and statistics, my work describes the functions a network can represent, the geometry that determines which solutions training selects, and the conditions under which a model’s internal structure can be identified from its behavior. The aim is a mathematical foundation for interpretability, reliability, and AI alignment: a precise account of which claims about learned systems can be proven.	
EMPLOYMENT & AFFILIATIONS	Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG) <i>Dresden, Germany</i> <i>Section of Mathematics and Artificial Intelligence</i> Postdoctoral researcher Mentor: Heather Harrington	2025–present
	ILIAD Fellow, applied mathematics for AI alignment	Sep–Dec 2026
EDUCATION	University of California, Los Angeles Ph.D. in Statistics, research focus in Machine Learning Theory Thesis: <i>Structure, Symmetry, and Singularity in Learning Systems</i> Thesis advisor: Guido Montúfar	2025
	The University of Hong Kong & Stony Brook University B.S. (Hons) Mathematics, Algebraic Geometry Thesis advisor: Robert Lazarsfeld	2018
HONORS, FELLOWSHIPS & GRANTS	ILIAD Fellowship, applied mathematics for AI alignment Conference grant, German Research Foundation (€4,000) Dimitris N. Chorafas Foundation Award (USD 10,000) Dissertation Year Fellowship, UCLA (USD 20,000) Distinguished Teaching Award Nominee, Statistics/MSOL, UCLA Summer Mentored Research Fellowship, UCLA (USD 6,000) ACM Scholarship, Association for Computing Machinery (USD 2,000) Cathay Bank Scholarship (USD 6,000) Overseas Research Fellowship, HKU (HKD 12,000) Undergraduate Research Fellowship, HKU (HKD 15,000) Travel grants: 20+ awards, over USD 25,000 cumulative	2026 2026 2024 2024 2024 2022 2020 2020 2016 2015 2017–2025
PREPRINTS	- Ben Cullen, Sergio Estan-Ruiz, Riya Danait, Jiayi Li. “A Basin-Selection Perspective on Grokking via Singular Learning Theory”, arXiv:2603.01192, under review - Cecilie Olesen Recke, Sarah Lumpp, Nataliia Kushnerchuk, Janike Oldekop, Jiayi Li, Jane Ivy Coons, Elina Robeva. “Identifiability in Graphical Discrete Lyapunov Models”, arXiv:2601.21818, minor revision at <i>Electronic Journal of Statistics</i>	
SELECTED PUBLICATIONS	- Yulia Alexandr, Miles Bakenhus, Maize Curiel, Sameer K. Deshpande, Elizabeth Gross, Yuqi Gu, Max Hill, Joseph Johnson, Bryson Kagy, Vishesh Karwa, Jiayi Li, Hanbaek Lyu, Sonja Petrović, Jose Israel Rodriguez. “New Directions in Algebraic Statistics: Three Challenges from 2023”, <i>Algebraic Statistics</i>, 15, 357–382, 2024 - Kaie Kubjas, Jiayi Li, Maximilian Wiesmann. “Geometry of Polynomial Neural Networks”, <i>Algebraic Statistics</i>, 15, 295–328, 2024	

Selected Publications (continued)

4. Shuang Liang, Renata Turkeš, **Jiayi Li**, Nina Otter, Guido Montúfar. “Pull-back Geometry of Persistent Homology Encodings”, Transactions on Machine Learning Research (TMLR), 2024
3. **Jiayi Li**, Yuantong Li, Xiaowu Dai. “Discussion of ‘Estimating Means of Bounded Random Variables by Betting’ by Waudby-Smith and Ramdas”, Journal of the Royal Statistical Society: Series B (JRSSB), 86(1), 41–43, 2023
2. Dejun Guo, Xu Jin, Dan Shao, **Jiayi Li**, Yang Shen, Huan Tan. “Image-Based Regulation of Mobile Robots without Pose Measurements”, IEEE Control Systems Letters (L-CSS), vol. 6, pp. 2156–2161, 2022
1. Ziqi Huang, Yang Shen, **Jiayi Li**, Marcel Fey, Christian Brecher. “A Survey on AI-Driven Digital Twins in Industry 4.0: Smart Manufacturing and Advanced Robotics”, Sensors, 21(19), 6340, 2021

LEADERSHIP &
MENTORING

- Co-organizer, Singular Learning Theory Days (two-day international workshop)
MPI-CBG, Dresden; funded by a DFG conference grant Oct 26–27, 2026
- Organizer, AMS Special Sessions at the Joint Mathematics Meetings 2025–2027
Algebraic Methods in Machine Learning and Optimization
JMM 2025, Seattle; JMM 2026, Washington D.C.
Algebraic, Topological and Geometric Methods in Machine Learning
JMM 2027, Chicago
- Minisymposium organizer, SIAM Conference on Mathematics of Data Science (MDS26)
Salt Lake City, UT, USA Nov 16–20, 2026
- Editor-in-Chief, XRDS: Crossroads, the ACM magazine for students (Association for
Computing Machinery) 2022–2025
Lead Editor 2021; Feature Editor 2020
- Co-mentor with Martin Trapp, LOGML 2026 Jul 13–17, 2026
Project: *Posterior Geometry and Variational Estimation of Bayesian Complexity*
- Mentor, LOGML 2025 Jul 7–11, 2025
Project: *Symmetry, Degeneracy and Effective Dimensions of Neural Networks*
The project developed into the preprint arXiv:2603.01192
- Academic mentor, IPAM Research in Industrial Projects for Students (RIPS)
Project: *Geometry of Representation in Large Language Models* Jun–Aug 2025
Industrial partner: Microsoft Research

RESEARCH VISITS

- Institute for Computational and Experimental Research in Mathematics (ICERM)**
Providence, RI, USA Jan 20–Apr 23, 2027 (scheduled)
In-residence participant, Semester Program: *Metric Algebraic Geometry — Going Global*
- Institute for Pure & Applied Mathematics (IPAM)**
Los Angeles, CA, USA Aug 31–Sep 4, 2026
Workshop participant (poster and lightning talk), *Foundations of Interpretability*
- Max Planck Institute for Mathematics in the Sciences (MPI MiS)**
Leipzig, Germany Summers 2024 and 2025
Visiting Graduate Researcher, Group: Mathematical Machine Learning
- Institute for Pure & Applied Mathematics (IPAM)**
Los Angeles, CA, USA Sep–Dec 2024
Graduate Fellow, Long Program: *Mathematics of Intelligences*

Simons Institute for the Theory of Computing

Berkeley, CA, USA

Short-Term Visitor, *Modern Paradigms in Generalization*

Fall 2024

Institute for Mathematical & Statistical Innovation (IMSI)

Chicago, IL, USA

Graduate Fellow, *Algebraic Statistics and Our Changing World*

Sep–Dec 2023

INVITED TALKS (24)

24. Machine Learning in Life Science (MLLS) Seminar

Oct 1, 2026, DTU, Lyngby, Denmark (scheduled)

“Structure, Symmetry and Singularity in Modern Neural Networks”

23. Mathematics Colloquium, Nanjing University

Aug 24, 2026, Nanjing, China

“Structure, Symmetry and Singularity in Modern Neural Networks”

22. Academy of Mathematics and Systems Science, Chinese Academy of Sciences

Aug 12, 2026, Beijing, China

“Structure, Symmetry and Singularity in Modern Neural Networks”

21. Machine Learning Seminar, École Normale Supérieure

May 21, 2026, Paris, France

“Structure, Symmetry and Singularity in Modern Neural Networks”

20. Conference on Geometry, Topology and Machine Learning, Sorbonne Université

May 18–22, 2026, Paris, France

“Basin Selection in Bayesian Learning”

19. Applied CATS Seminar, KTH Royal Institute of Technology

Mar 3, 2026, Stockholm, Sweden

“Grokking as a Phase Transition Between Competing Basins, an SLT Perspective”

18. Mathematical Foundations of AI, Lie–Størmer Colloquium

Dec 3–5, 2025, Simula Research Laboratory, Oslo, Norway

“Structure, Symmetry and Singularity in Learning Systems”

17. Math & Machine Learning Seminar, Boston College

Oct 15, 2025, online

“Theoretical insights for optimizing rational activations”

16. Applied Algebra and Geometry UK Network, Edinburgh meeting

Aug 21, 2025, Edinburgh, UK

“Geometry of Neural Networks with Algebraic Activations”

15. Algebraic Methods in Optimization, ICCOPT

Jul 21–24, 2025, Los Angeles, USA

“Optimizing rational neural networks with trainable parameters”

14. SIAM Conference on Applied Algebraic Geometry

Jul 7–11, 2025, Madison, WI, USA

“Geometry of Neural Networks with Algebraic Activations”

13. Mathematics, AI and Data Science for Material Innovations (MADSMIN), Lancaster University

Jun 9–13, 2025, Lancaster, UK

“Geometry of Neural Networks with Algebraic Activations”

- 12. Geometry and Machine Learning (GML2025), Sorbonne Université**
May 26–30, 2025, Paris, France
“Geometry of Neural Networks with Algebraic Activations”
- 11. Workshop on the Statistical Theory of Neural Networks**
May 5–8, 2025, Egmond aan Zee, Netherlands
“Geometry of Neural Networks with Algebraic Activations”
- 10. Numerical (Nonlinear) Algebra in the Real World**
Feb 6, 2025, Dresden, Germany
“Taming Non-convexity in Shallow Neural Networks with Algebraic Activations”
- 9. Joint Mathematics Meetings (JMM) 2025**
AMS–ASA–SIAM Special Session on Mathematics of Deep Learning: A High-Dimensional Probability Perspective
Jan 11, 2025, Seattle, WA, USA
“Algebro-geometric Approaches to Optimization and Generalization in Mathematical Machine Learning”
- 8. Algebraic Geometry and Machine Learning Workshop, SIAM Conference on Mathematics of Data Science**
Oct 25, 2024, Atlanta, GA, USA
“Geometry of Polynomial Neural Networks”
- 7. Numerical Analysis Seminar, Institute of Mathematical Research and Dept. of Mathematics, The University of Hong Kong**
Apr 30, 2024, Hong Kong
“Geometry of Polynomial Neural Networks”
- 6. Special Session: Applications of Algebra and Geometry, American Mathematical Society 2024 Spring Central Sectional Meeting**
Apr 20, 2024, Milwaukee, WI, USA
“An Algebraic Approach to Supply Network Formation and Fragility”
- 5. Dept. of Mathematics and Statistics, University of Massachusetts Amherst**
Apr 12, 2024, online
“Geometry of Polynomial Neural Networks”
- 4. Level Set Seminar, Dept. of Mathematics, UCLA**
Apr 8, 2024, Los Angeles, CA, USA
“Geometry of Polynomial Neural Networks”
- 3. Applied Algebra Seminar, Dept. of Mathematics, University of Wisconsin–Madison**
Dec 2023, Madison, WI, USA
“Geometry of Polynomial Neural Networks”
- 2. Deep Learning Theory Seminar, UCLA**
Sep 2023, Los Angeles, CA, USA
“Neural Collapse Beyond Unconstrained Feature Models”
- 1. SCISS, UCLA**
Jul 2023, Los Angeles, CA, USA
“Mathematical Machine Learning: Theory & Beyond”

CONTRIBUTED TALKS
(6)

6. Workshop on Real Applied Algebra

Mar 23–25, 2026, University of Copenhagen, Denmark

“LLC in transformers: An algebro-geometric lens on generalization in neural networks”

5. Higher-order interactions at the crossroads of geometry and topology

Dec 10–12, 2025, Paris, France

“Geometry of Higher-Order Structures in Machine Learning”

4. Matroids, Rigidity, and Algebraic Statistics, ICERM Workshop

Mar 17, 2025, Providence, RI, USA

“Geometric Framework for Optimizing Algebraic Neural Networks”

3. 2024 UC Davis Peter Hall Conference on Statistics in the Age of AI

Nov 9, 2024, Davis, CA, USA

“Optimization in Polynomial Neural Networks: Insights from Algebraic Geometry”

2. Women in Algebraic Statistics Workshop, St John’s College, University of Oxford

Jul 12, 2024, Oxford, UK

“Algebraic Structures in Terminal Phase of Neural Network Training”

1. Institute for Mathematical and Statistical Innovation (IMSI)

Oct 10, 2023, Chicago, IL, USA

“Simplex ETF – Algebraic structure arising in the terminal phase of neural network training”

TEACHING

Summary: Instructor, Teaching Associate, and Teaching Assistant for 25+ course offerings across statistics, engineering, and machine learning, reaching 1000+ students at UCLA and the University of Oxford.

Instructor of record: Prep Camp, Master of Applied Statistics & Data Science (MASDS), UCLA, 2021–2023

Teaching Associate / Teaching Assistant, UCLA, 2020–2023:

Statistics and Data Science (STATS 13, ENGR 116)

Engineering Systems (ENGR 200, 202, 203, 205)

Teaching Assistant, Oxford Machine Learning Summer School

SERVICE &
OUTREACH

Outreach

Founding organizer, Distinguished Women in Statistics and Data Science Workshop Series, UCLA: 1st (Women in Academia, May 23, 2023) and 2nd Symposium (Women in Industry, Oct 17, 2023)

UCLA Society of Women in Statistics (Committee, 2022–2024)

UCLA Statistics Club (Mentor, 2022–2024)

Invited speaker: Rotary Club, Westwood Village (2023); UCLA Bruin Professionals (2024); Dublin High School, Dublin, CA (2024)

Reviewing

NeurIPS, ICML, ICLR, Algebraic Statistics, SIAM Journal on Applied Algebra and Geometry, Effective Methods in Algebraic Geometry (MEGA)

MEMBERSHIPS

American Mathematical Society (AMS), Association for Women in Mathematics (AWM), American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), Society for Industrial and Applied Mathematics (SIAM), Association for Computing Machinery (ACM)